

Time-Varying Effects of Health Technology on Racial Disparities: Why When We Measure Matters

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SER 2026 | *Methods and Statistics: Advancing Racial and Socioeconomic Equity in Treatment and Health Outcomes*

Disclosures/Conflicts of Interest

- Work completed during my PhD program
- Views expressed in this presentation represent my own, and do not represent the views of the Henry M. Jackson Foundation or the Uniformed Services University of the Health Sciences
- No relevant financial relationship(s) exist

Consider Two Examples...

Do wearable cardiac monitors reduce the gender gap in timely arrhythmia detection?

Does access to a free home blood pressure monitor widen the income gradient in hypertension control?

Neither were designed to address social issues, yet they may contribute to a change in disparities.

Relevant Questions about Health Technology

Can a technology *widen* a disparity even when it helps everyone?

- Net benefit and equity are not the same metric.

And (of course) are we measuring the right thing?

- Disparities evaluation asks, "for whom, under what conditions, and at what point in the care pathway?"
- If we only estimate average effects, we miss the structural mechanisms that determine who benefits most

Three Ways Technology Widens Disparities

ACCESS

Who can use it



Telemedicine¹

EFFICACY

How well it works



Pulse oximeter²

DOWNSTREAM

What it reshapes

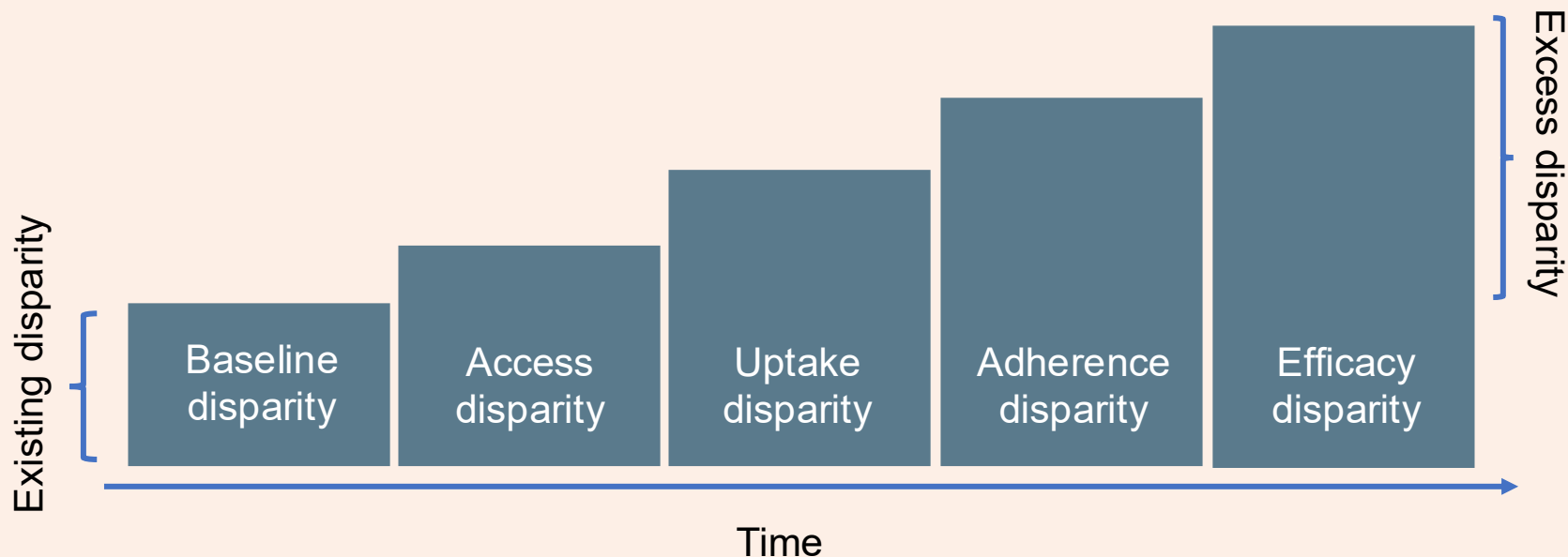


AI data centers³

How Does Health-Tech Affect Health Disparities?

Intervention-generated inequalities (IGI)⁴:

technological interventions work better for those who are already better off



Motivating Example: Type 2 Diabetes

CDC National Diabetes Statistics Report, 2026; CDC/NCHS NHANES; HHS Office of Minority Health

Type 1 (5–10%)

Autoimmune destruction of insulin-producing cells. Requires insulin.

Type 2 (90–95%)

Insulin resistance plus relative insulin deficiency. Heavily managed by lifestyle changes and meds.

A large and unequal burden

1 in 8

U.S. adults have diabetes

21% vs 11%

prevalence among non-Hispanic Black vs. 11% in White adults

25%

US hospitalizations due to diabetes; significantly higher for Black people with T2D

Motivating Example: Type 2 Diabetes

Type 2 is socially and structurally determined

Yet, primary prevention and treatment focuses heavily on lifestyle and behavioral changes

Determinants of Type 2 Diabetes

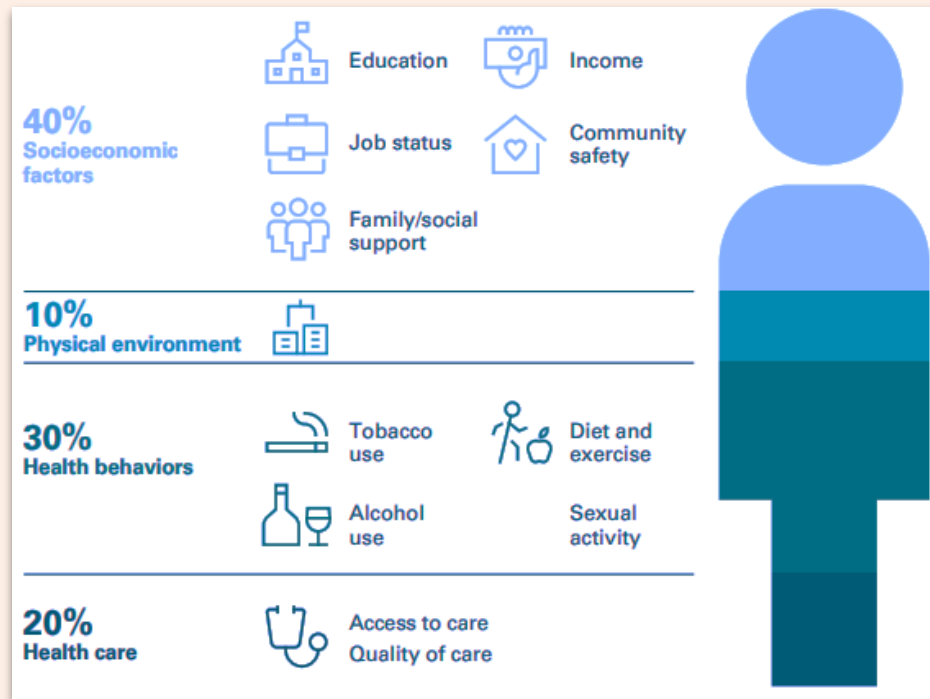


Image: BCBS Minnesota, n.d.

Continuous Glucose Monitors

CGMs monitor your blood glucose continuously and provide alerts when the blood glucose is too high or low⁵

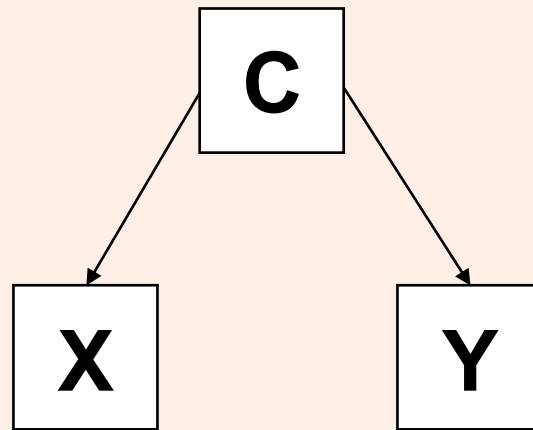
Can encourage the behavior changes which improve blood sugar control⁶

Who benefits most? When?



Causal Question

Does the White-Black racial disparity in hospitalization increase, decrease, or remain unchanged when a population with T2D initiates the use of a CGM compared to if they do not initiate a CGM?



Target Trial Emulation

<i>Components</i>	<i>Target trial</i>	<i>Emulation of target trial</i>
Eligibility criteria	Age 18+, has a diagnosis of Type 2 Diabetes, no prior use of a CGM	All of Us Research Program participants with self-reported type 2 diabetes, no use of a CGM in prior 6 months, have A1c lab results within last 3 months
Treatment strategies	CGM use vs. normal diabetes management/care	Same. Operationalized as initiation by CPT codes indicative of CGM set up or data interpretation
Treatment assignment	Random assignment; will be aware of their assignment	Same as target trial
Follow-up	After treatment assignment through end of study, death, or loss to follow up, whichever occurs first	Same as target trial
Outcome	Time until first hospitalization	Same at target trial. Operationalized as any hospitalization indicated by CPT codes
Analysis	Time to event analysis to assess hospitalizations. Inverse probability of treatment weighting (IPTW) To adjust for potential selection bias due to loss to follow-up.	Same as target trial. IPTW to adjust for confounding. A pooled sequence of trials where eligibility is assessed on a rolling monthly basis.

Sample

2,376

Black
participants

8,983

White
participants

1.7%

Black
CGM use

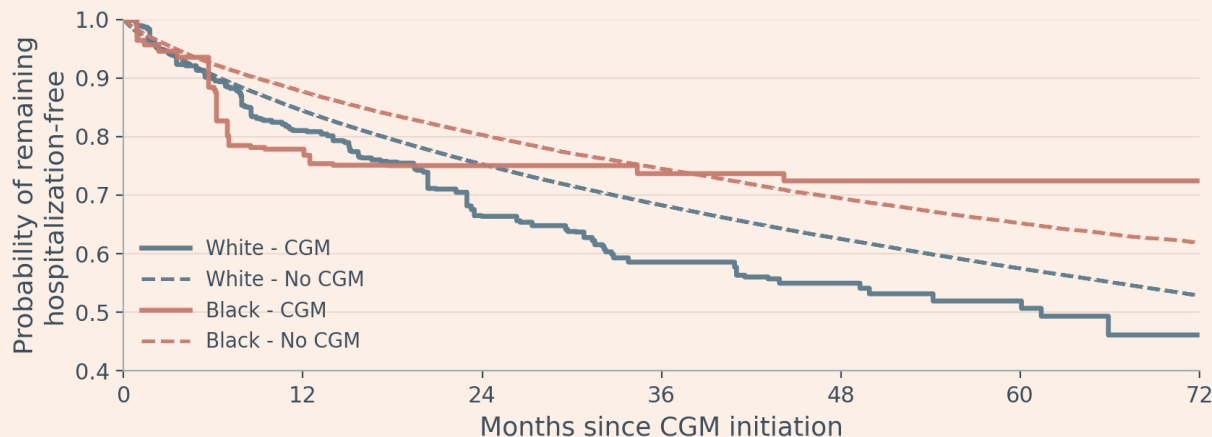
3.5%

White
CGM use

Characteristic	Black	White
Household income $\leq$ \$35k	50%	25%
College graduate	28%	49%
High neighborhood deprivation	59%	25%
Mean A1c (%)	7.00	6.80
Age $\geq$ 65	20%	39%

Hazard Ratio

Time to first hospitalization, by race and CGM use
(higher = longer hospitalization-free survival)



White vs. Black
under treatment

HR = 1.88

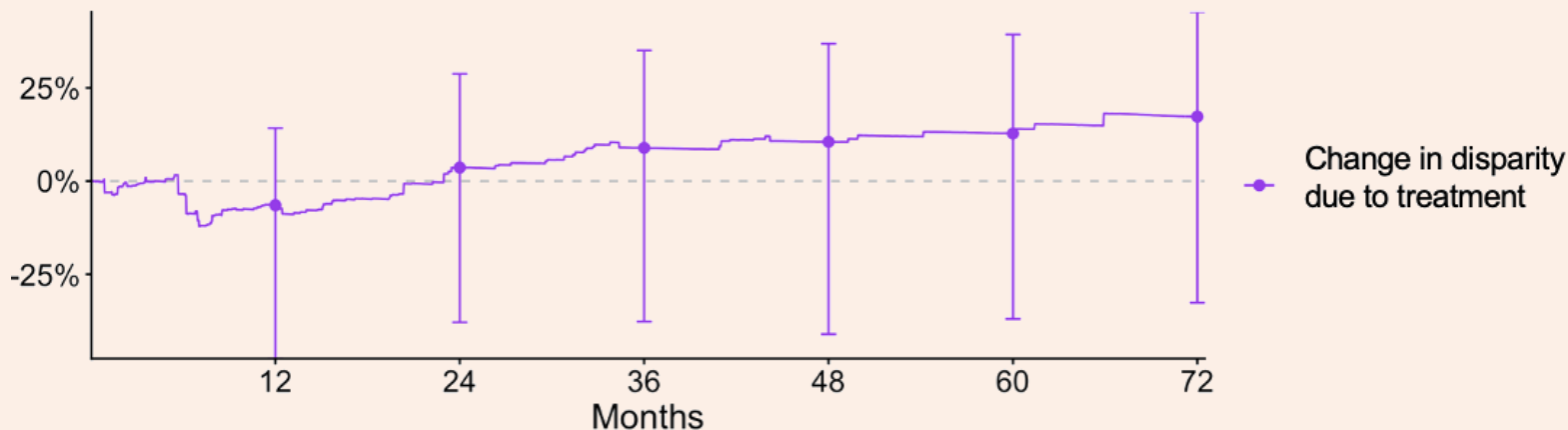
95% CI: 0.33, 8.34

Among treated,
averaged across
follow-up, implies
White risk is
steadily higher

Ah, the Hazards of Hazard Ratios!⁷

CGM Effect on Disparities Flips After 6 Months*

Change in White-Black hospitalization disparity due to CGM use
(positive = White higher risk, negative = Black higher risk)



-0.10

12 months
Black at higher risk

+0.01

24 months
Groups converge

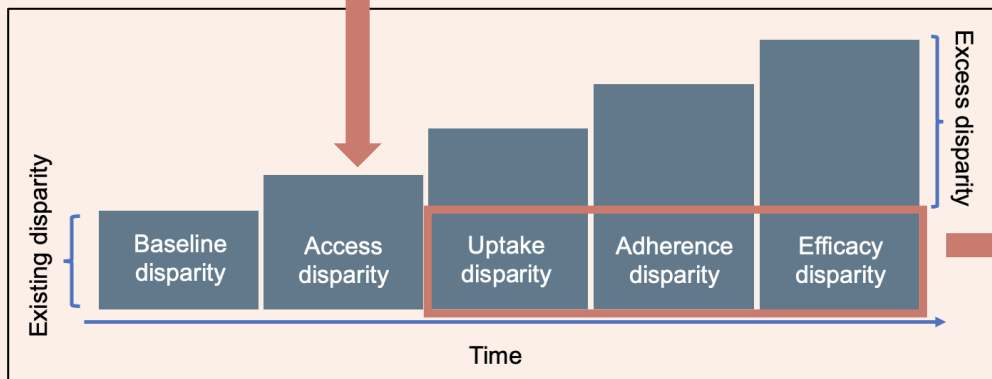
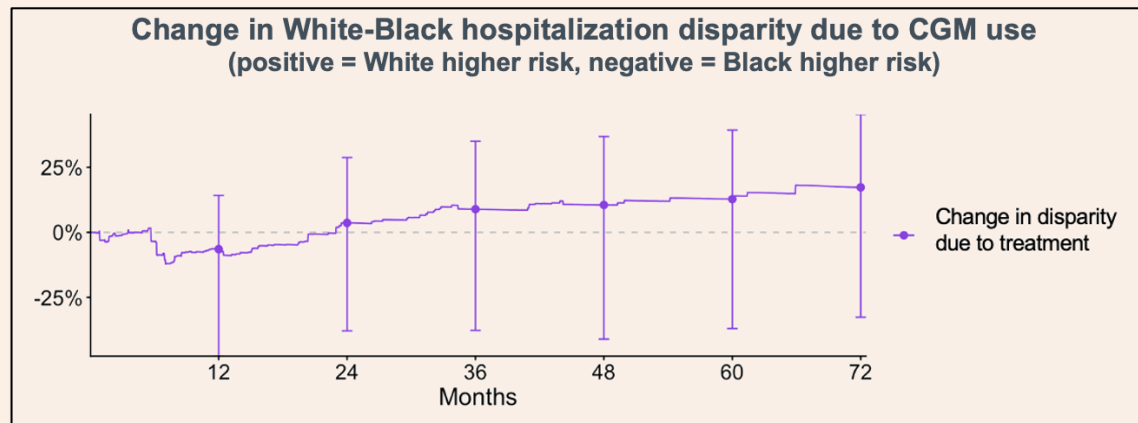
+0.11

72 months
Black at lower risk

* yes, the confidence intervals are huge and include the null!

Intervention-Generated Inequalities

Intervention: equalized
access at baseline



Differences in uptake,
adherence, efficiency
over time

Intervention-Generated Inequalities

One hypothesis might be:

- **Early period** → Black initiators clear a higher acuity threshold → sicker at baseline (higher baseline A1c) → higher early risk
(This is the disparity, captured by design.)
- **Late period** → White initiators persist passively; differential post-initiation persistence lets sicker White users accumulate person-time → higher long-term risk

Final Note - Low Sample Size and Wide CIs

Problem: low sample size leads to wide confidence intervals

Low n among Black CGM users (n=41) is not just a study limitation

- Direct consequence of the inequity under study
- Represents the differential technology uptake
 - Which is part of the disparity



Final Note - Low Sample Size and Wide CIs

So, what do we do?

Sparse data in the disadvantaged group

+ summary measure

= findings highly sensitive to the follow-up window chosen

Track the direction of effects over follow-up
(rather than a precise summary)

Why When We Measure Matters

1

Technology effects are not static

User engagement evolves as individuals learn from devices; the devices themselves change via software updates

2

Structural factors shape long-term benefits differentially

Early health tech use may prompt healthcare-seeking; sustained benefits depend on things like food access, care quality, and neighborhood resources, which are all socially patterned; Technology $\neq$ structural change

3

Summary hazard ratios collapse temporal variation

Well-established critiques of summary statistics (i.e., crude odds ratio, hazard ratio) are amplified when both engagement patterns and structural barriers change over follow-up (Greenland, 1996; Hernán, 2010)

4

Single time-point evaluations risk premature conclusions

Can be misleading about about *who* benefits from a technology and *when*

Teaser!

Combining Target Trial Emulation with Causal Decomposition to Estimate Intervention Effects on Health Disparities

A Framework and Application

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SER 2026 | *Pathways Shaping Diabetes Risk and Health Disparities*

Thank you!

All of Us Research Program
participants

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Questions?

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Thank you!

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